

MA540 Project 1

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Problem 1

Statement

We consider the spring model:

$$\begin{aligned}\frac{d^2 z}{dt^2} + C \frac{dz}{dt} + Kz &= 0, \\ z(0) &= 2, \quad \frac{dz}{dt}(0) = -C,\end{aligned}$$

with inputs $\theta = [K, C]$. To compute the sensitivities $\frac{\partial z(t)}{\partial K}$ and $\frac{\partial z(t)}{\partial C}$, we will use complex-step approximation with $h = 10^{-20}$. To do so, we replace C with $C + ih$, compute $z(t)$ using an ODE solver, take the imaginary part, and divide by h , approximating the derivative. We repeat for K . We will also approximate the sensitivities using forward finite-difference approximation, with $h = 10^{-5}$ in the following formula:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

using $K = 20.5$ and $C = 1.5$. We also compare with the analytical solutions:

$$\begin{aligned}\frac{\partial y}{\partial K} &= e^{-Ct/2} \frac{-2t}{\sqrt{4K - C^2}} \sin\left(\sqrt{K - C^2/4} \cdot t\right), \\ \frac{\partial y}{\partial C} &= e^{-Ct/2} \left[\frac{Ct}{\sqrt{4K - C^2}} \sin\left(\sqrt{K - C^2/4} \cdot t\right) - t \cos\left(\sqrt{K - C^2/4} \cdot t\right) \right].\end{aligned}$$

Solution

We graph each over an interval of $t = 0$ to $t = 12$ at 500 equally spaced time-values. Below are the results comparing the sensitivities of each parameter (Figure 1 and Figure 2).

As seen below, both complex-step and forward difference approximate the analytical solutions to high precision. While we expect forward difference to be less accurate than complex-step, we see that difference is negligible in this situation. Both models appear to be valid approximations.

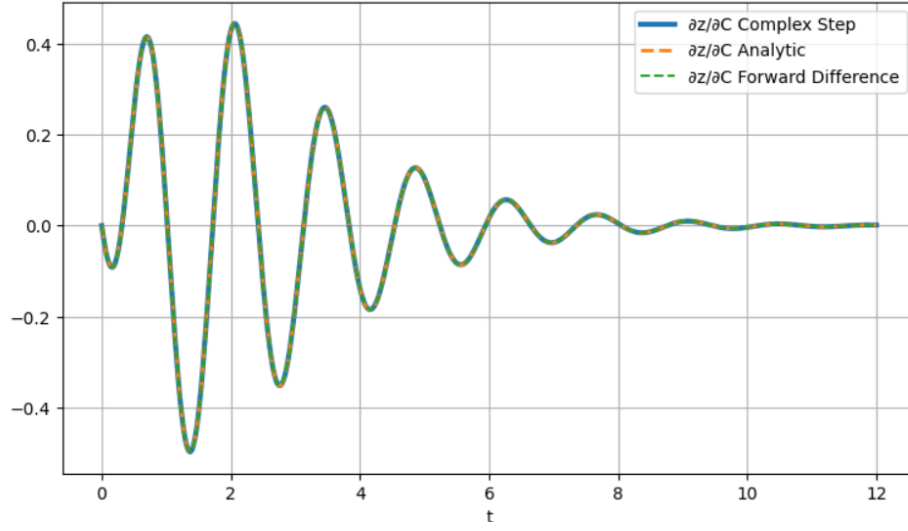


Figure 1: Sensitivities of C for Complex Step, Analytic, and Forward Difference Models

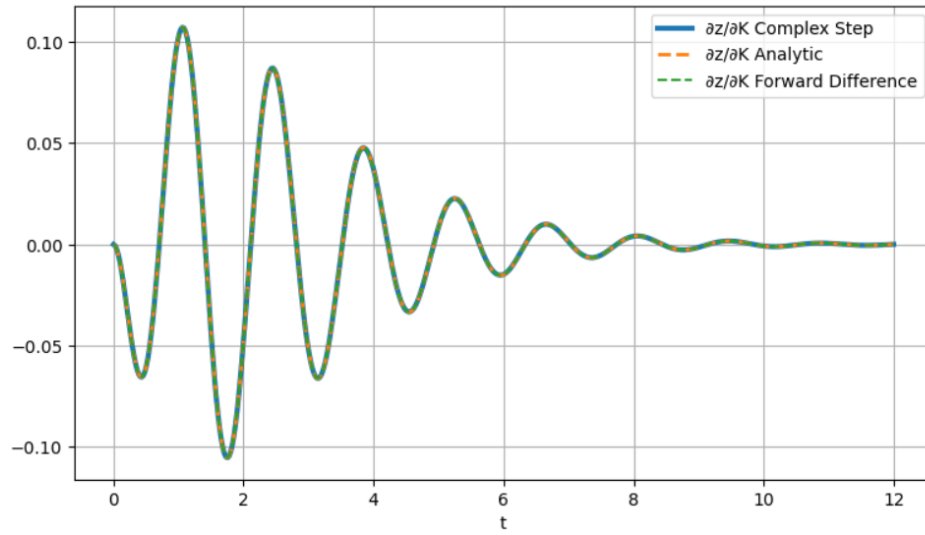


Figure 2: Sensitivities of K for Complex Step, Analytic, and Forward Difference Models

Problem 2

Statement

We consider the 4-parameter SIR model:

$$\begin{aligned}\frac{dS}{dt} &= \delta N - \delta S - \gamma k I S \quad , \quad S(0) = S_0, \\ \frac{dI}{dt} &= \gamma k I S - (r + \delta) I \quad , \quad I(0) = I_0, \\ \frac{dR}{dt} &= r I - \delta R \quad , \quad R(0) = R_0,\end{aligned}$$

with initial conditions $S_0 = 900$, $R_0 = 0$ and $I_0 = 100$. We numerically solve this ODE for $R(t)$, and employ an algorithm to test the sensitivity of each parameter. We shift each parameter by $h = 10^{-5}$, numerically solve the ODE for $R(t_i)$, and use forward-difference approximation to calculate the sensitivity across 51 equally-spaced time values of $t_i \in [0, 5]$. This is formulated into a sensitivity matrix, following the form:

$$S(\boldsymbol{\theta}) = \frac{\partial \mathbf{y}}{\partial \boldsymbol{\theta}} = \begin{bmatrix} \frac{\partial y_1}{\partial \theta_1} & \frac{\partial y_1}{\partial \theta_2} & \cdots & \frac{\partial y_1}{\partial \theta_n} \\ \frac{\partial y_2}{\partial \theta_1} & \frac{\partial y_2}{\partial \theta_2} & \cdots & \frac{\partial y_2}{\partial \theta_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial \theta_1} & \frac{\partial y_m}{\partial \theta_2} & \cdots & \frac{\partial y_m}{\partial \theta_n} \end{bmatrix}.$$

for $y = R(t_i)$, $\boldsymbol{\theta} = [\gamma, k, \delta, r]$, and respective nominal values $[0.2, 0.1, 0.15, 0.6]$. We then perform singular value decomposition, and look for the parameters with singular values less than our tolerance of $\varepsilon = 10^{-2}$ to determine an identifiable parameter set.

Solution

Below is the graphed solutions of the SIR system:

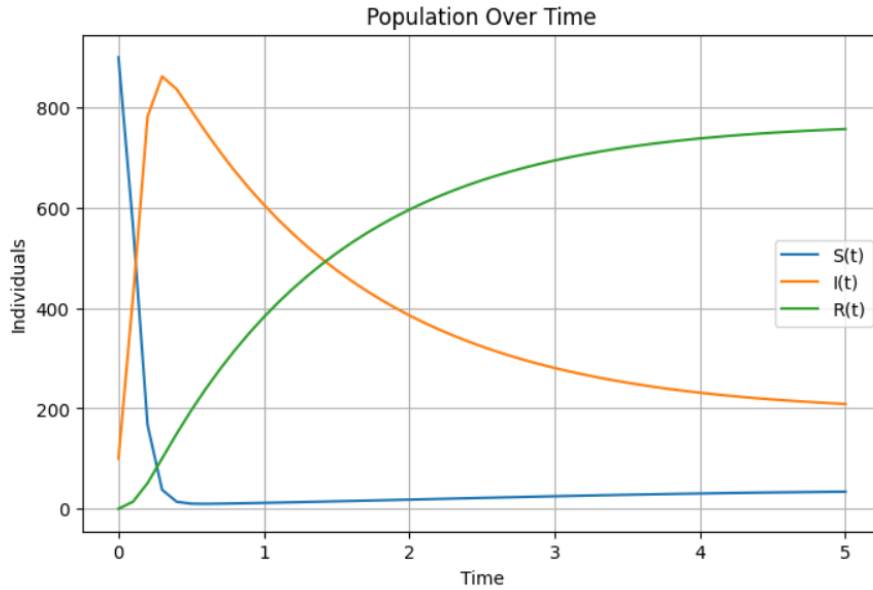


Figure 3: Solution Set of the SIR Model via Numerical ODE Solver

We find the identifiable parameter set to be $\boldsymbol{\theta} = [\gamma, k, \delta]$. This is because the singular value for $r = 8.19 \times 10^{-3} < \varepsilon = 10^{-2}$. By comparison, the singular values for the identifiable parameters are much larger: $\delta = 5.74 \times 10^3, \gamma = 1.99 \times 10^3, k = 8.12 \times 10^2$.

Problem 3

Statement

We consider the boundary value problem

$$\begin{aligned} \frac{d^2 T_s}{dx^2} &= \frac{2(a+b)}{ab} \frac{h}{k} [T_s(x) - T_{amb}] \\ \frac{dT_s}{dx}(0) &= \frac{\Phi}{k} \quad , \quad \frac{dT_s}{dx}(L) = \frac{h}{k} [T_{amb} - T_s(L)] \end{aligned}$$

where $T_s(x)$ denotes the steady state temperature of an uninsulated rod with source heat flux Φ at $x = 0$ and ambient air temperature $T_{amb} = 21.29^\circ\text{C}$. The rod has cross-sectional dimensions $a = b = 0.95$ cm and length $L = 70$ cm. The model parameters are $\boldsymbol{\theta} = [\Phi, h, k]$, with nominal values $[-18.4, 0.00191, 2.37]$. First, we will use forward finite difference and complex-step approximation to compute the sensitivities $\frac{\partial f}{\partial \Phi}$, $\frac{\partial f}{\partial h}$ and $\frac{\partial f}{\partial k}$, as detailed in Problem 1. We will compare these approximations to the analytic solutions

$$f(x, \boldsymbol{\theta}) = T_s(x, \boldsymbol{\theta}) = c_1(\boldsymbol{\theta})e^{-\gamma x} + c_2(\boldsymbol{\theta})e^{\gamma x} + T_{amb},$$

where $\gamma = \sqrt{\frac{2(a+b)h}{abk}}$ and

$$c_1(\boldsymbol{\theta}) = -\frac{\Phi}{k\gamma} \left[\frac{e^{\gamma L}(h + k\gamma)}{e^{-\gamma L}(h - k\gamma) + e^{\gamma L}(h + k\gamma)} \right] \quad , \quad c_2(\boldsymbol{\theta}) = \frac{\Phi}{k\gamma} + c_1(\boldsymbol{\theta}).$$

Next, we will construct the sensitivity matrix S_{ij} by shifting each parameter by $h = 10^{-20}$ and computing the sensitivity relation by employing complex-step approximation on the given analytic solutions. We then compute the singular values of S . We will compare these to our tolerance of $\varepsilon = 10^{-3}$ to determine identifiability. Finally, we will fix $k = 2.37$ and repeat the sensitivity matrix analysis without varying k , again calculating singular values and eigenvalues to determine identifiability.

Solution

Below are the graphs of the sensitivities $\frac{\partial f}{\partial \Phi}$, $\frac{\partial f}{\partial h}$ and $\frac{\partial f}{\partial k}$, plotted over $x_i \in [10, 66]$ at 14 equally-spaced x -values.

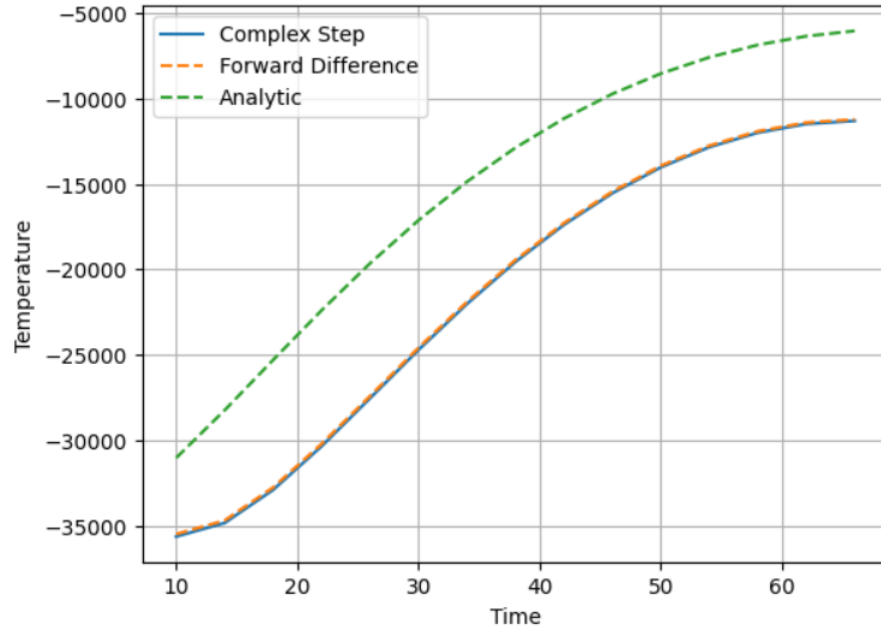


Figure 4: Sensitivities of h for Complex Step, Analytic, and Forward Difference Models

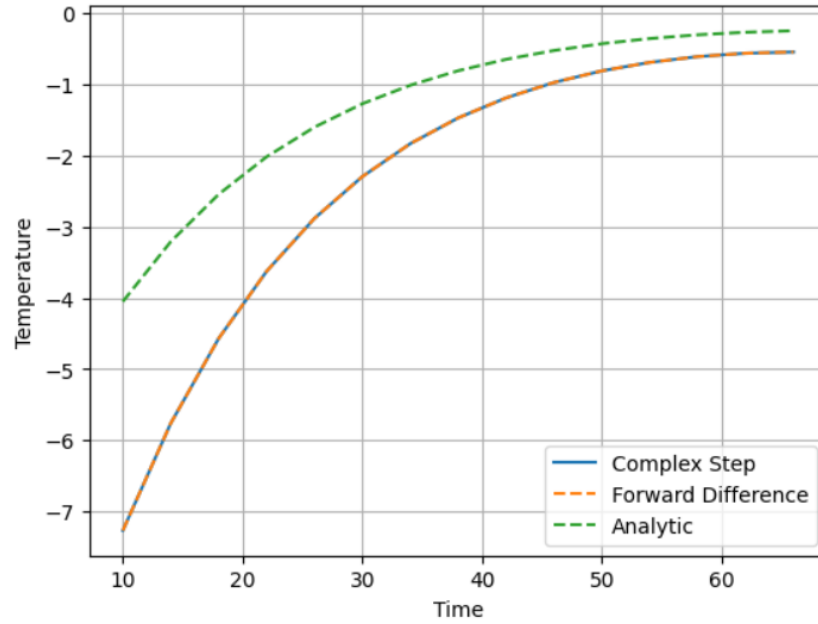


Figure 5: Sensitivities of Φ for Complex Step, Analytic, and Forward Difference Models

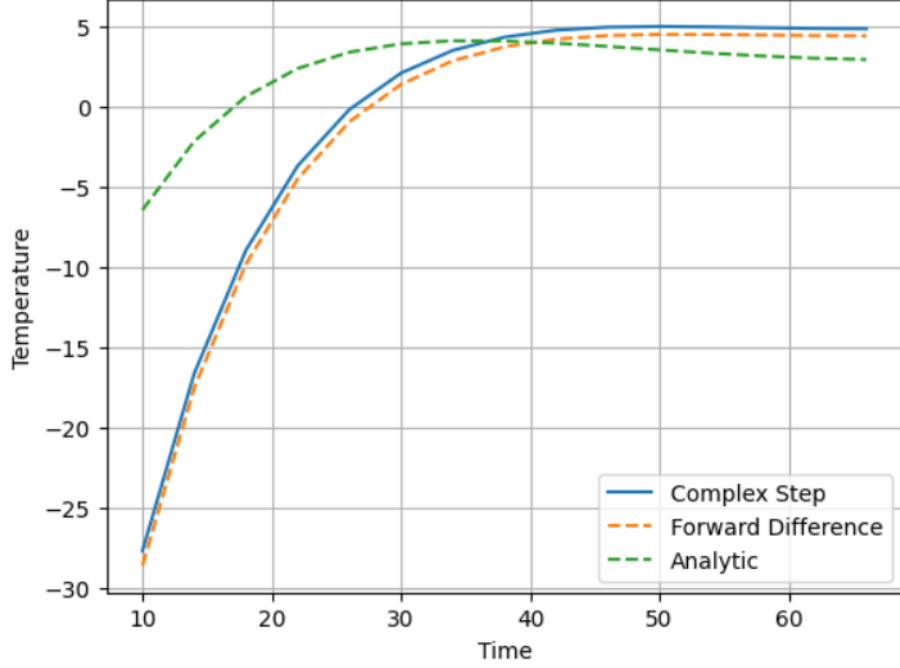


Figure 6: Sensitivities of k for Complex Step, Analytic, and Forward Difference Models

As shown, the approximations deviate slightly from the analytical solutions, but approximate the shape well. The similarity of the complex-step and forward difference approximations is unexpected. We compute the singular values of the sensitivity matrix S to be $[6.67 \times 10^4, 1.35 \times 10^1, 1.25 \times 10^{-13}]$, respective to $[h, k, \Phi]$. Hence, Φ is less than our tolerance, and determined to be unidentifiable. This is consistent with the model parametrization, as Φ only appears in quotients with k , and never in a separate linear combination. Repeating our analysis with a fixed k -value, we find the singular values to be $[6.67 \times 10^4, 1.73 \times 10]$. Hence, both parameters are now identifiable. As predicted, fixing the parameter that Φ appears in a quotient with allowed Φ to become identifiable.

Problem 4

Statement

We consider the Helmholtz energy

$$\psi(P, \boldsymbol{\theta}) = \alpha_1 P^2 + \alpha_{11} P^4 + \alpha_{111} P^6,$$

where P is the polarization and $\boldsymbol{\theta} = [\alpha_1, \alpha_{11}, \alpha_{111}]$ are parameters, with nominal values $\alpha_1 = -389.4$, $\alpha_{11} = 761.3$ and $\alpha_{111} = 61.5$. First, we will plot the energy on the interval $P \in [-0.8, 0.8]$ by the nominal values. Next, we will compute the sensitivity matrix S by shifting each parameter by $h = 10^{-20}$ and finding the respective sensitivity relation with complex-step approximation. This is done over $x_i \in [0, 0.8]$ at 17 equally-spaced x -values. We will then compute the singular values of S and discuss identifiability with $\varepsilon = 10^{-1}$. Thirdly, we will use Morris screening with forward differences and $r = 50$, $\Delta = 1/20$, sampling each parameter from $\mathcal{U}(0, 1)$ and mapping to the interval $[\alpha_\ell, \alpha_r]$ that is 20% above and below the nominal value to compute μ_i^* and σ_i^2 . We do this by

inputting the randomly sampled values into the scalar computed analytically by

$$y(\theta) = \int_0^{0.8} \psi(P, \theta) dP, \quad (1)$$

and shifting one parameter by Δ in the form of forward difference, as shown in Problem 1. We then calculate μ and σ^2 for each parameter.

Solution

Below is the graph of energy on the interval $P \in [-0.8, 0.8]$:

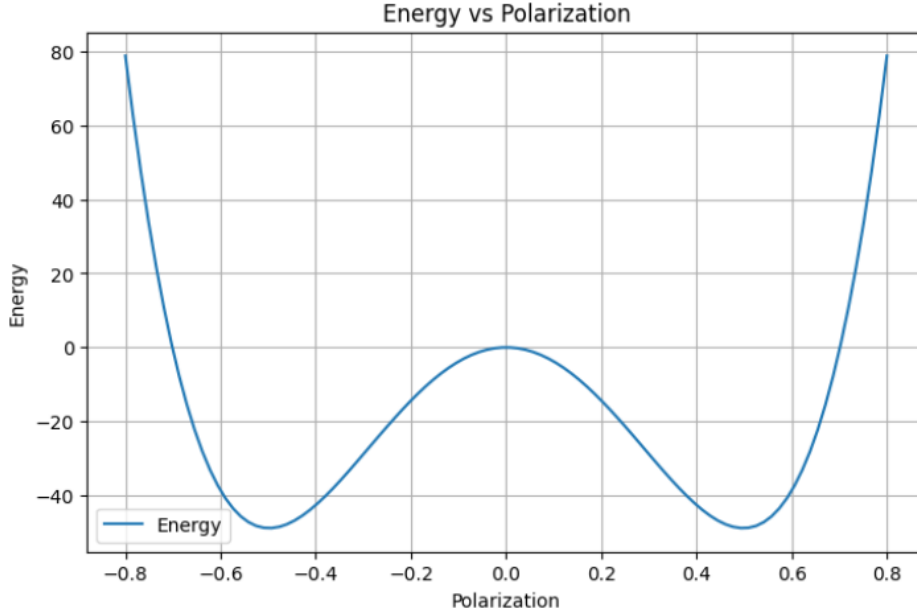


Figure 7: Energy vs Polarization

As expected from a Helmholtz energy equation, the graph exhibits double-well behavior as an even function. We find the singular values of the sensitivity matrix to be $[1.41, 2.06 \times 10^{-1}, 2.07 \times 10^{-2}]$, respective to $[\alpha_1, \alpha_{11}, \alpha_{111}]$. Hence, α_{111} is unidentifiable, which is expected for the sixth-order term in an energy equation. The lower-order terms have a much greater effect on the shape of the graph, while the sixth-order term offers very minimal advantage in model accuracy. We find the global sensitivity values for each parameter to be as follows:

$$\alpha_1 : \mu = 10.92, \sigma^2 = 11627.82$$

$$\alpha_{11} : \mu = -890.20, \sigma^2 = 11627.82$$

$$\alpha_{111} : \mu = -891.23, \sigma^2 = 11515.22$$

This solution does not appear to follow the expected values, suggesting the analysis performed is not fit for the situation.